



This introduction to Hadoop will tell you how to install and configure it in Windows. The prerequisites, the associated software and other requirements are all listed here, and there's advice on the dos and don'ts for a successful installation.

Hadoop is an Apache open source software library written completely in Java, designed to deliver a distributed file system (HDFS) and a method for distributed computation called MapReduce. It can scale up from single servers to thousands of machines.

It provides a framework that allows distributed processing of large sets of data across clusters of computers using simple programming models.

Hadoop implements MapReduce, which runs in two phases: the map phase and the reduce phase. The input to these computations is the data set of key/pair values. In short, the application is divided into many small fragments, each of which may be assigned to the map task. These map tasks are distributed and executed on any node in the cluster. If the nodes fail during the computation process, the tasks assigned to them are redistributed among the remaining nodes.

The Apache Hadoop project includes the following modules:

Hadoop Common: The common utilities that support the other Hadoop modules.

Hadoop Distributed File System (HDFS): A distributed file system that provides high-throughput access to application data.

Hadoop Yarn: A framework for job scheduling and cluster resource management.

Hadoop MapReduce: A yarn-based system for parallel processing of large data sets.

Hadoop deployment methods

Here is a list of the methods used to deploy Hadoop.

Standalone: In this mode, there are no daemons running; everything runs as in a single JVM. This mode is suitable for running the MapReduce program during development, as it is easy to test and debug.

Pseudo-distributed: The Hadoop daemon process runs on a local machine simulating a cluster on a small scale.

Fully distributed: Here, Hadoop runs on a cluster of machines providing a production environment.

This article focuses on introducing Hadoop, and deploying single-node pseudo-distributed Hadoop on a Windows platform.

Prerequisite software or tools for running Hadoop on Windows

You will need the following software to run Hadoop on Windows.

Supported Windows OSs: Hadoop supports Windows Server 2008 and Windows Server 2008 R2, Windows Vista and Windows 7. For installation purposes we are going to make use of Windows 7 Edition and JDK.

As Hadoop is written in Java, we will need to install Oracle JDK 1.6 or higher. Download Java from the link given in References [1] at the end of this article, and install it to the default location.